

1 PP-PDC integration in HYDRA

Purpose

If the ERP system and HYDRA are connected via the PP-PDC interface, the subsystem must be able to receive the [IDocs](#) generated by SAP ECC and to integrate them in the HYDRA process. HYDRA also generates IDocs from the recorded downloads/confirmations and transfers them to SAP ECC. SAP ECC triggers both workflows.

SAP provides several standard BAPIs / IDocs as part of the PP-PDC interface to establish communication with BDE subsystems. The following BAPIs / IDocs are used:

Download operations:

IDoc type:	PPCC2RECORDER01
Segment type:	E1BP_PP_PDC_OPERA2
BAPI segment:	BAPI_PP_PDC_OPERA2

Request uploads:

IDoc type:	PPCC2REQCONF
Segment type:	E1PPCC2REQCONF
BAPI segment:	BAPI_PP_PDC_PARAM

Upload time tickets:

IDoc type:	PPCC2PRETTICKET01
Segment type:	E1BP_PP_TIMETICKET
BAPI segment:	BAPI_PP_TIMETICKET

Unlike the previous interface, the communication channel 2, the new PP-PDC interface is completely based on BAPI and RFC. This means that the SAP transceiver is no longer required and that new RFC servers and clients are implemented in HYDRA. In HYDRA these new components communicate with SAP ECC and constitute the interface itself.

SAP ECC → HYDRA (download operation data)

HYDRA RFC server

The HYDRA RFC server logs in to the SAP gateway. The RFC server receives the incoming IDocs and stores them to the HYDRA database. Then the HYDRA process responsible for transferring data to the HYDRA data model is started.

BAPI / IDoc types

SAP ECC uses SAP workflow processes to trigger the download of data to HYDRA. Basically, the standard PP-PDC interface provides three IDoc structures. The HYDRA RFC server receives these IDoc structures: OPERA2 (initial / delta download), OPERA1 (deletion requests) and REQCONF (confirmation/upload requests).

Operation data (OPERA2)

An IDoc of the type PPCC2RECORDER01 transfers the operations. Depending on the entry in the segment E2PPCC2RECORDER000, it is either an initial download or a delta download. The segment E2BP_PP_PDC_OPERA2 includes operation data.

Deletion download (OPERA1)

An IDoc of the type PPCC2RECORDER01 transfers the operations. When it comes to a delta download, the segment E2BP_PP_PDC_OPERA1 can transfer the keys of the operations that are to be deleted.

Upload request (REQCONF)

An IDoc of the type PPCC2REQCONF transfers the request to upload time tickets. If this IDoc type is received, it is an upload request.

	OPERA2		OPERA1	REQCONF
	Initial download	Delta download	Deletion download	Upload request
IDoc type	PPCC2RECORDER01	PPCC2RECORDER01	PPCC2RECORDER01	PPCC2REQCONF01
Segment name	E2PPCC2RECORDER	E2PPCC2RECORDE R	E2PPCC2RECORDE R	E2PPCC2REQCONF
Field	INIT	INIT	INIT	REQTT
Entry	X			X

HYDRA → SAP ECC (upload)

SAP ECC asks for the upload of time tickets (L20 / L40). As an option, HYDRA can also start the upload at regular intervals, irrespective of an upload request.

HYDRA RFC client

The HYDRA RFC client uploads the time tickets. The client is started as part of a HYDRA workflow. At this time, the data is available ready for dispatch in the database.

Data is transferred asynchronously as IDoc. Once transferred, a defined SAP ECC workflow processes and posts the data.

Time tickets

An IDoc of the type PPCC2PRETTICKET01 of the standard PP-PDC interface transfers the time tickets. The segment E2BP_PP_TIMETICKET includes the time ticket data.

In general, the following SAP time ticket record types are uploaded:

- L20 – Partial confirmation (reporting part quantities/times)
- L40 – Final confirmation (reporting total quantities/times)